# BEFORE vs AFTER
# FOR EACH ROW vs FOR EACH STATEMENT

Consider the relation

*Flight(passengerName, airline, flightNumber, date, flightMiles)*

Write a trigger that resets the miles to 300 for any passenger-flight for which there is an attempt to insert or to change the miles to less than 300.

- Write the trigger using both the before and after options
- Write the trigger using both options for the granularity levels (for each statement or for each row)

# BEFORE vs AFTER - FLIGHT MILES

```
create trigger 300Miles
before insert on Flight
for each row
when new.flightMiles<300
begin
  set new.flightMiles=300;
end;
```

```
create trigger 300Miles
before update of flightMiles on Flight
for each row
when new.flightMiles<300
begin
  set new.flightMiles=300;
end;
```

# BEFORE vs AFTER - FLIGHT MILES

```
create trigger 300Miles
after insert on Flight
for each row
when new.flightMiles<300
begin
  update Flight
  set flightMiles=300
  where
        passengerName=new.passengerName
        and flightNumber=new.flightNumber
        and date=new.date
end;
```

```
create trigger 300Miles
after update of flightMiles on Flight
for each row
when new.flightMiles<300
begin
  update Flight
  set flightMiles=300
  where
        passengerName=new.passengerName
        and flightNumber=new.flightNumber
        and date=new.date
end;
```

(NOT POSSIBLE IN SOME DBMS!)

# FOR EACH STATEMENT -FLIGHT MILES

```
create trigger 300Miles
after insert on Flight
for each statement
REFERENCING NEW TABLE AS NEW_T
begin
  update Flight
  set flightMiles=300
  where (passengerName,flightNumber,date)
          IN (select passengerName,
                        flightNumber, date
              from NEW_T
              where flightMiles<300)
end;
```

```
create trigger 300Miles
after update of flightMiles on Flight
for each statement
REFERENCING NEW TABLE AS NEW_T
begin
  update Flight
  set flightMiles=300
  where (passengerName,flightNumber,date)
          IN (select passengerName,
                        lightNumber, date
              from NEW_T
              where flightMiles<300)
end;
```

(NOT POSSIBLE IN SOME DBMS!)